ELEC 221: Lab Report Guidelines

- Each individual who is doing the lab **must bring a pre-lab report** to the lab. The pre-lab reports will be **initialized by the TA**.

- Each group must submit a **complete Lab Report (one PDF file per group)** to the OnQ Dropbox.

Section	When to Complete	Notes
Lab report	During and after the lab	Record all experimental steps, results, and observations as a group while performing in the lab. After completing the lab, work together to finish analysis, discussion, and conclusions. **Only one shared lab report is required**, which should include the Experiment section with results.

Note: The final completed version of the Lab Report must be submitted to the OnQ Dropbox by the deadline announced on OnQ.

Recommendation: We strongly encourage all group members to read the lab manual before coming to the lab. Having an overview of the procedure in advance will help your group follow the steps more smoothly and understand the purpose behind each part of the experiment.

ELEC221: Electric Circuits
Lab 3 Pre-Lab: Linear Circuit Elements

ELEC 221- Fall 2026 **Lab 3 Report**

Group Number:	
Date Performed:	
Section Number:	

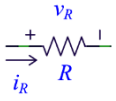
Group Members Name	Group Members Student Number

Objectives:

1. Study the behavior and step response of **linear circuit elements** (RC and RL circuits)
2. Verify the Thevenin equivalent of a given circuit.

Linear Components:
There are three types of linear components found in electrical circuits; resistors, capacitors, and inductors. These are all passive elements. That is, they do not generate energy whose voltage–current relationship is linear.

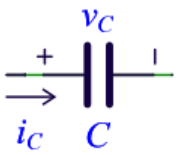
The Resistor: A resistor is a linear circuit element that impedes the flow of current. The relationship between current and voltage for a resistor is $\boldsymbol{v = iR}$.



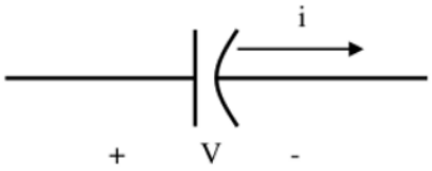
The Capacitor: A capacitor is a linear circuit element that stores energy in an electric field. The relationship between current and voltage for a capacitor is: $i_C = C\frac{dv_C}{dt}$

The graphic symbol for a capacitor indicates that this component is built from two conducting parallel plates separated by a dielectric. Under DC conditions, current cannot flow from one plate

to another. Therefore, by Ohm's law, the resistance of an ideal capacitor is infinite. We can easily build a capacitor using aluminum foil as the conductive plates and wax paper as the dielectric.



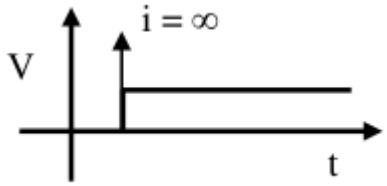
or



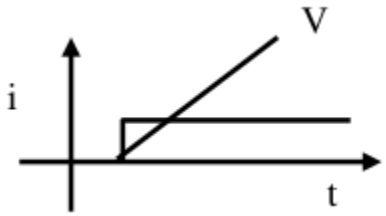
From the equation of the current through a capacitor, we can identify three limiting cases for ideal capacitors.

DC Steady State: If the voltage across the capacitor is constant, there will be no current flowing through the capacitor. In other words, the capacitor acts as an open circuit.

Voltage Step: If we apply a voltage step to a capacitor, the capacitor will conduct an infinite current spike.

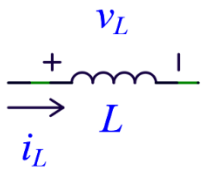


Current Step: If a current step is applied to the capacitor, the voltage across the terminals of the capacitor will increase linearly with time.



The Inductor is a linear circuit component that stores energy in a magnetic field. The relationship between current and voltage for an inductor is: $v_L = L\frac{di_L}{dt}$

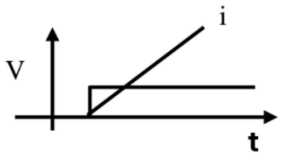
A simple form of inductor is a wire coil. We can build an inductor by simply wrapping a long wire around a pen. Since the inductor is simply a long wire, the resistance should be small - for an ideal inductor, we assume it to be zero.



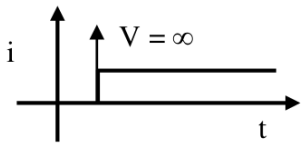
Similarly, as we did for the capacitor, we can identify three limiting cases for voltage across an ideal inductor.

DC Steady State: If a constant current is flowing through the inductor, there will be no voltage drop across the terminals of the inductor. The inductor acts as a short circuit.

Voltage Step: If a voltage step is applied to the terminals of an inductor, the current through the inductor will increase linearly with time.



Current Step: If we apply a current step to the inductor, it will cause an infinite voltage spike.



Thevenin Equivalent

In this section of the lab, we will analyze the maximum power transfer of a purely resistive circuit and compare it to the maximum power transfer of the Thevenin equivalent of the circuit.

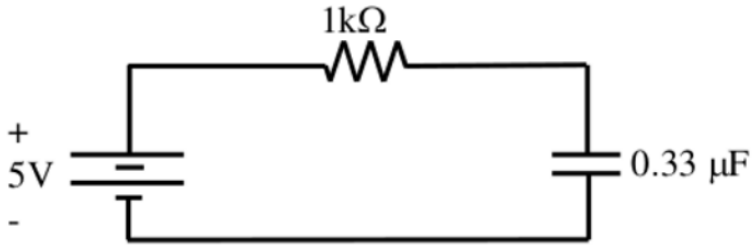
Thevenin Equivalent: The simplification of a circuit containing sources and resistors into an equivalent circuit composed of one voltage source connected in series with one resistor.

Maximum Power Transfer: The condition that maximizes the power that can be delivered to a load.

Preparation:

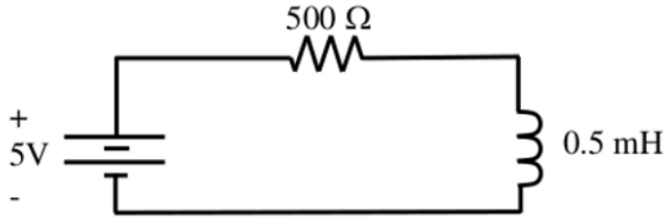
Linear Circuit Components

1. Direct current (DC) means something that is constant in time. A 5-volt DC source is connected in series with a $0.33\mu F$ capacitor and a $1K\Omega$ resistor.
   a) The voltage across the capacitor (V_C)
   b) The voltage across the resistor (V_R)



$V_C =$_____
$V_R =$_____

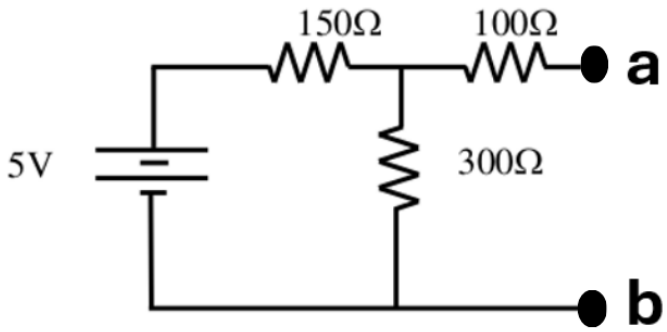
2. A 5-volt DC source is connected in series with an ideal $0.5\ H$ inductor and a 500Ω resistor. After the circuit has stabilized, what is the current in the circuit? What is the voltage across the resistor?



$V_R =$_____
$I =$ _____

Thevenin Equivalent

3. Across the terminals A and B, the circuit on the left has a Thevenin equivalent circuit shown on the right.



Calculate the open-circuit voltage (V_{OC}) , short-circuit current (I_{SC}) and R_{th}:

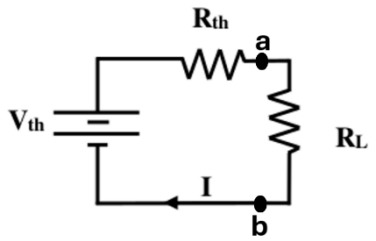
V_{OC}= _____

I_{SC}=_____

R_{th} =_____

When a load resistor is connected on terminals A and B, current flows, and power is dissipated in the load.

The Thevenin equivalent circuit from question 3 with a load resistor ($R_L = R_{Load}$), added to terminals **'a'** and **'b'**. The current through the resistors is: $I = \frac{V_{th}}{R_{th}+R_L}$.



The **power** across a load is given by the formula: $P_L = V_L I_L$. The derivatives of the power formula can be obtained by using **ohm's law** ($V = I \times R$).

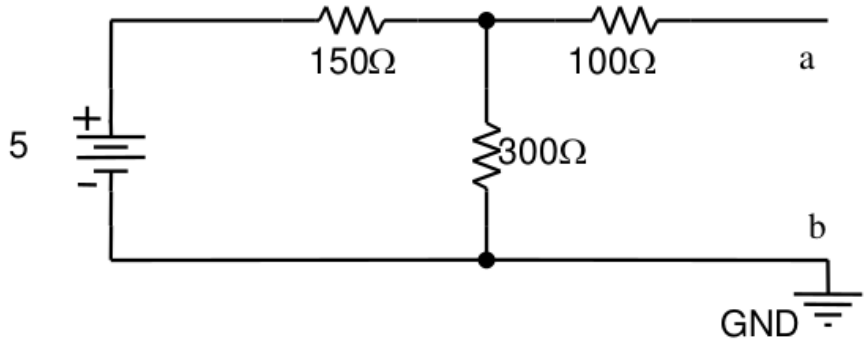
4. By combining the above formula, ($I = \frac{V_{th}}{R_{th}+R_L}$, $P_L = V_L \times I_L$, $V = I \times R$), express the power dissipated in the load, P_L, as a function of R_L.

5. Calculate the value of R_L for which P_L is a maximum.
   For a given value of R_{th}, the power delivered to load has a maximum when $R_{Load} = R_{th}$.

LTspice Exercise:

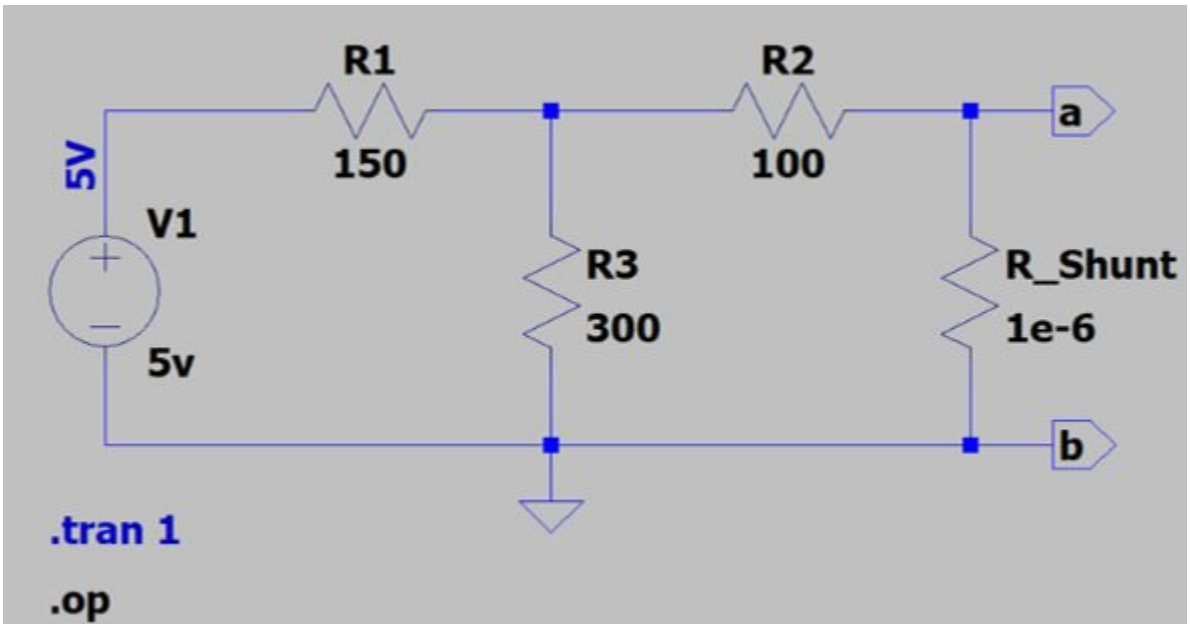
Although LTspice cannot give you the Thevenin equivalent of a circuit at given terminals, it can help you find the **open circuit voltage** and the **short circuit current** at a pair of terminals.
A) Open the LTspice and construct the following design (For installation and

setup instructions, you can use the Appendix from the Lab 2 report).



From the **Simulation** tab in LTspice, select **Configure Analysis**, then choose **"DC op pnt"** tab. In the syntax box, enter ".op". Run the simulation and review the **Visible Traces** chart (results).

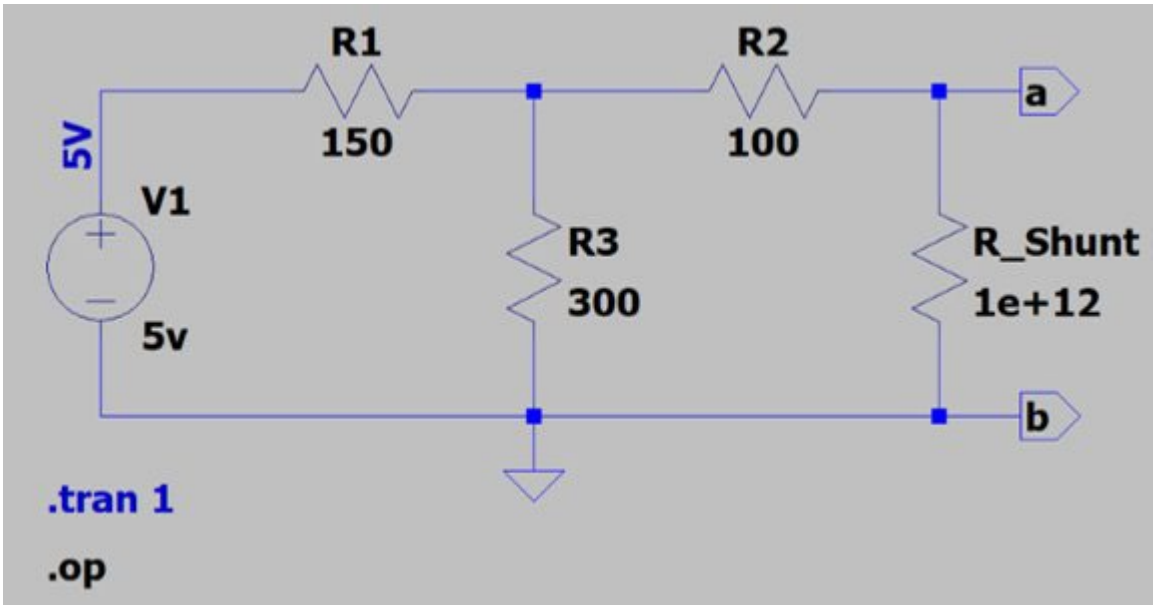
Find the voltage drop across the $1\mu\Omega$ resistor and calculate the current going through the resistor. This is the short circuit I_{SC}.



B) Compare this value to the value of I_{SC} you found in question 3 of the prelab.

$V_{1\mu\Omega}$ =_____	I_{SC}=_____

C) Change the value of the $1\mu\Omega$ resistor to $1T\ \Omega$ Run the simulation again.
This time, the voltage across the $1T\ \Omega$ resistor will give you the **open circuit voltage**, V_{OC}.



Compare this value with the value of V_{OC} you found in question 3 of the prelab.

V_{OC} =_____

ELEC221: Electric Circuits
Lab 3 (Experiments): Linear Circuit Elements

Procedure:

Note: There are more comprehensive questions in the *Analysis Section* of the lab report that deal with understanding each part of the experiment. We recommend that your **group** read through the Analysis Section and make sure you understand the questions before moving on to the next part. However, do not try to answer all the questions during the lab session, unless time permits.

Part A: Linear Circuit Elements

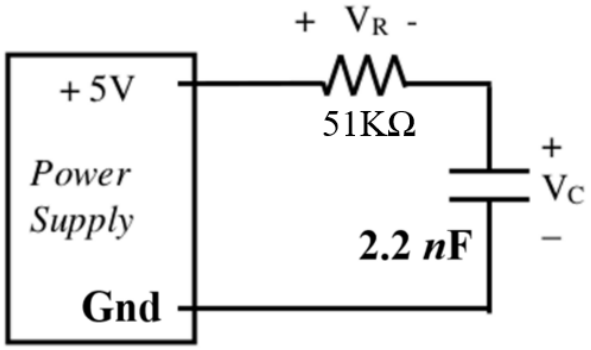
1. A) With the capacitor, select a $2.2\ nF$ capacitor and measure its resistance with the digital meter.

$R_C =$_____

B) Take the $100\ mH$ inductor and measure its resistance with the digital meter.

$R_L =$_____

2. Assemble the circuit shown below using the components provided. Set the capacitance to $2.2\ nF$ and the resistance to $50\ K\Omega$. Using the digital voltmeter, measure the voltage across the resistor (V_R) as well as the voltage across the capacitor (V_C).

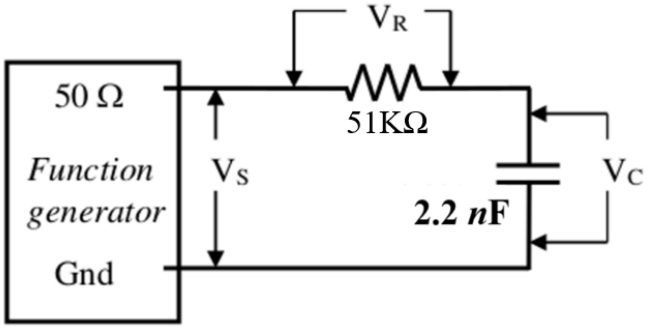


Resistor	Capacitor	V_R	V_C
50 $K\Omega$	2.2 nF		

3. Replace the capacitance from the circuit with an 100 mH inductor and select 1 $K\Omega$ on the resistor. Measure and record the voltage across both the resistor and the inductor.

Resistor	Inductor	V_R	V_L
1 $K\Omega$	100 mH		

4. Set the **function generator** to generate a 5 V **pk-pk** square wave with a 900 Hz frequency. Select a positive DC offset so that the signal goes from 0V to 5V . Construct the circuit below with the resistor and the capacitor.

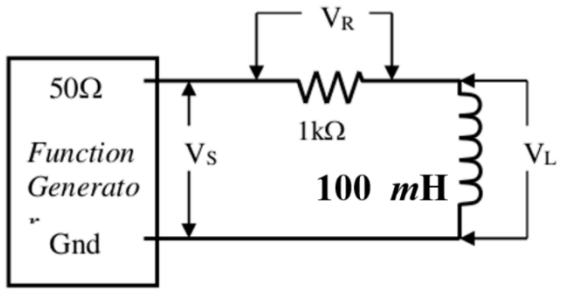


5. With the oscilloscope, use **channel 1** to measure V_S and **channel 2** to measure V_C . Select DC coupling on both channels to view the DC offset of the function generator.

Display V_S and V_C simultaneously on the **CH1** and **CH2** traces of the oscilloscope, lining them up by using a common zero voltage line on the screen. Using the horizontal scale knob on the oscilloscope. *Obtain a view of approximately two wavelengths. Sketch the waves below. Indicate the scales.*

6. Simultaneously display the input voltage, the voltage across the resistor and the voltage across the capacitor. Sketch the three waves below. Label each voltage curve, indicating the scales.

7. Using the inductor provided and the resistor, construct the circuit Below.

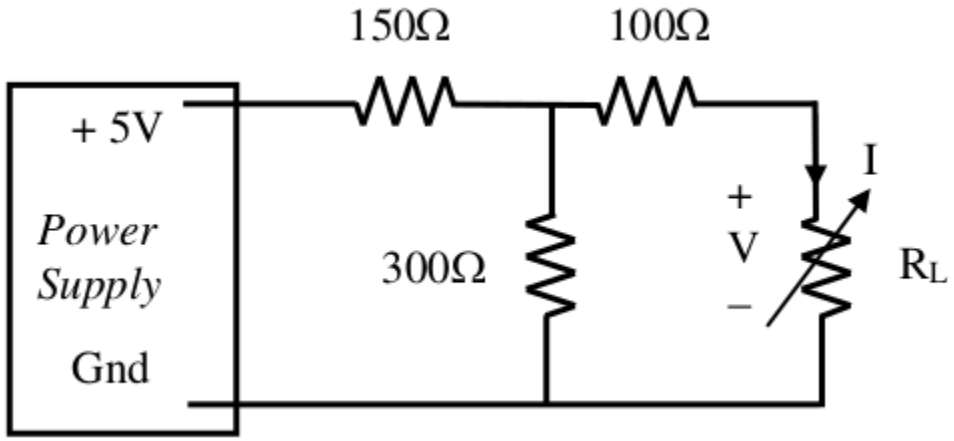


8. Measure V_S with **CH1** of the oscilloscope and V_L with **CH2**. Line up both waves to the common zero line on the oscilloscope screen. Obtain a clear view of approximately two wavelengths and sketch the input wave and inductor voltage wave.

9. Similarly, display the input voltage, the voltage across the resistor and the voltage across the inductor onto the screen of the oscilloscope and sketch the result.

Part B: Thevenin Equivalent

1. Construct the circuit shown below. Use a resistor for the variable resistance



2. Change R_L to get the values found in the table below. Measure the voltage across R_L and record the voltages in the second column. Calculate the current flow through the resistor as well as the power dissipated by the resistor. Place your results in the table below. ***Hint, for*** $R_L = 0\ \Omega$ ***to find the current through the load you can measure the voltage across the*** $100\ \Omega$ ***, and calculate the current.***

Resistance R_L	Voltage **V**	Current **I**	Power P_L
~ 0Ω			
50Ω			
100Ω			
150Ω			
200Ω			
500Ω			
1000Ω			
∞ (open)			

.

3. *Set the positive DC variable power supply on the breadboard to V_{OC} (value calculated* in question 3 of the Preparation Section) with the digital multimeter. Replace the circuit used above with its Thevenin equivalent circuit. Place the decade resistor as the R_L Load. Choose three resistance values from the table above for the load resistance values and fill in the table below.

Resistance R_L	*Voltage (V)*	*Current (I)*	*Power P_L*

Now that you have completed the Experiment section, please work with your group to answer the following questions.

These follow up questions are designed to help you analyze and reflect on your experimental results.

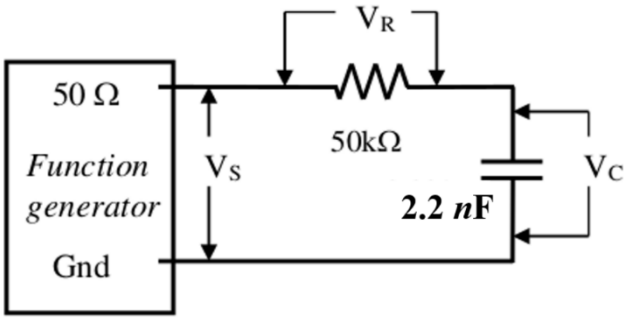
Part A: Linear Components (Analysis)

1. Compare the resistance of the capacitor measured in the lab to the value that may be inferred from part 1 of the preparation.

2. Compare the resistance of the inductor measured in the lab to the value that may be inferred from part 2 of the preparation. Explain why the values are not similar.

3. Using the sketch of the voltage across the capacitor in **Part A-5** of the Procedure Section, indicate the areas where:
   A) Energy is being stored in the capacitor.
   B) Energy is being delivered to the resistor by the capacitor.
   C) The capacitor has a maximum charge (i.e. No more energy can be stored into the capacitor.)

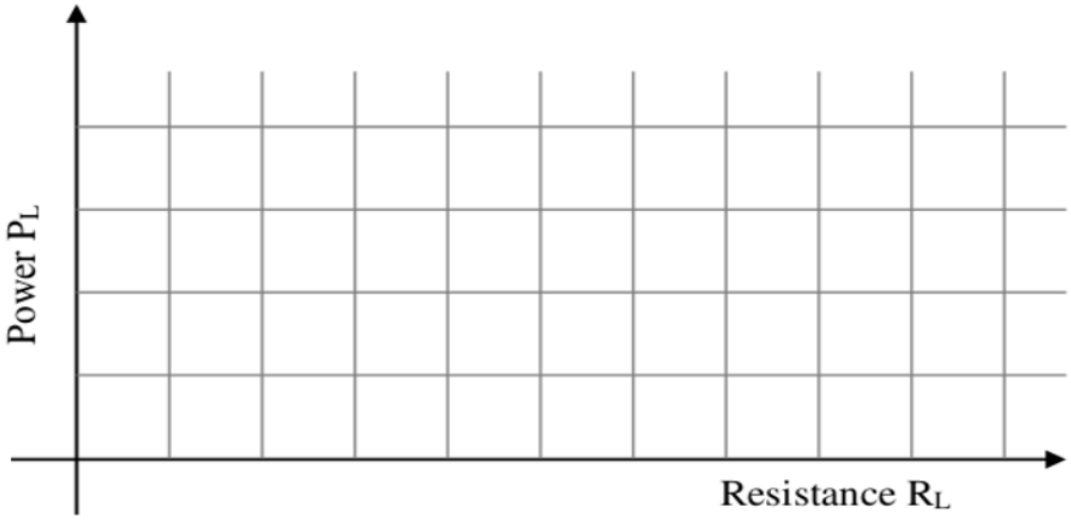
4. Does the circuit **bellow** reach a point where the voltages have the same values as exist in DC steady state? Explain why or why not.



5. Indicate on the graph of the voltage across the inductor (Part A-8 of the Procedure Section) the areas where:
   A) The inductor is acting as a pure resistance.
   B) The inductor is storing energy.

Part B: Thevenin Equivalent (Analysis)

1. With the table from the lab (Part B-2), plot the curve relating power to resistance in the graph provided below.



2.

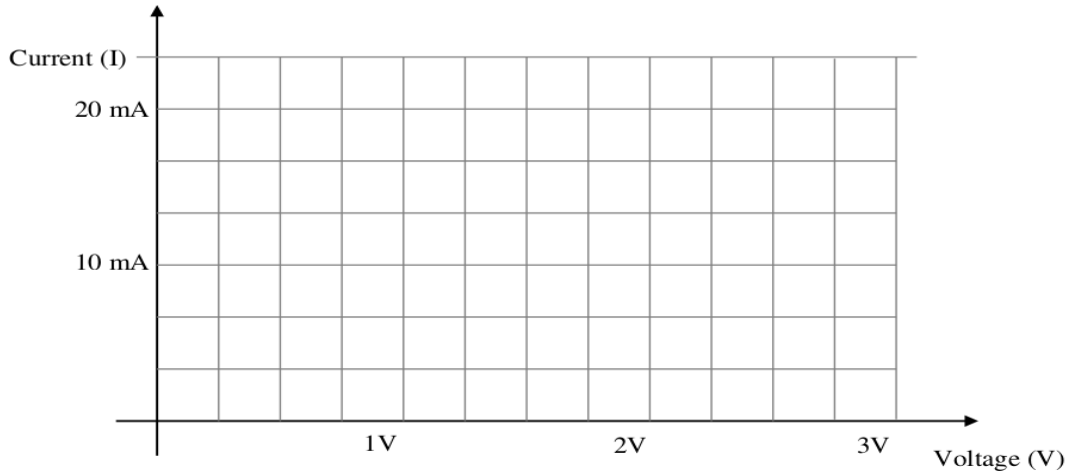
A) In the graph above, does the power delivered to R L exhibit a maximum? If so, estimate the value of R L for which P L is maximized.

R_L= _____

B) How does this value of R L relate to R th found in question 3 of the prelab?

3.

A) With the table from the lab (Part B-2) plot the load line. The definition of the load line is I versus V.



B) In the plot of the load line above, is the line linear? Explain why it is or isn't straight.